Newton's Universal Law of Gravitation

Name: ______________________________ **Date:** ________________

Part A: Concept Review

Gravitational Fields and Non-Contact Forces

- Every object with mass produces a **gravitational field**.

- Gravity is a **non-contact force** that acts at a distance.

- Gravitational fields extend outward in all directions.

Universal Law of Gravitation

The formula for the universal law of gravitation is provided by the following equation:

$$F_g = \frac{Gm_1m_2}{r^2}$$

- F_g is the magnitude of the gravitational force.
- m_1, m_2 are the masses of the objects.
- r is the distance between the centers of mass.
- $G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$, also known as the gravitational constant.
- The force decreases with the **square of the distance**.

What's the difference? G vs. g

- G is the universal gravitational constant.
- g is the local gravitational acceleration.
- Near Earth's surface, $g \approx 9.81\ \mathrm{m/s^2}$.

Let's derive the value of g

What is the gravitational force of an object of mass m near the surface of the Earth?

What is the gravitational force, using the universal law of gravitation?

Let's add some numbers to this:

- $G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$
- Mass of Earth: $M_E = 5.97 \times 10^{24}$ kg
- Radius of Earth: $R_E = 6.37 \times 10^{6}$ m

What is my value for g now?

Gravity and Orbital Motion

How does this relate to centripetal motion?

For circular orbits, gravity provides the centripetal force:

$$F_c = \frac{mv^2}{r}$$

$$\frac{mv^2}{r} = \frac{GMm}{r^2}$$

How can we solve for v?

Part B: Reference Data

- $G = 6.67 \times 10^{-11}\ \mathrm{N} \cdot \mathrm{m}^2/\mathrm{kg}^2$
- Mass of Earth: $M_E = 5.97 \times 10^{24}$ kg
- Radius of Earth: $R_E = 6.37 \times 10^6$ m
- Mass of Sun: $M_S = 1.99 \times 10^{30}$ kg
- Mass of Moon $M_m = 7.347 \times 10^{22}$ kg
- Distance from Earth to Moon 3.844×10^7m

Practice Problems

1. A 70 kg person stands on Earth's surface.

(a) Use $F = \frac{GM_E m}{R_E^2}$ to calculate the gravitational force on the person.

(b) Use $F = mg$ to calculate the person's weight.

(c) Explain why these two values are approximately equal.

2. (a) State the magnitude of the gravitational force the person exerts on Earth.

(b) State the direction of this force.

3. Using Newton's Second Law, calculate the acceleration of Earth caused by the person.

$$a_E = \frac{F}{M_E}$$

4. Two objects experience a gravitational force F at distance r.

(a) What happens to the force if the distance doubles?

(b) What happens to the force if the distance triples?

(c) By what factor must the distance change for the force to become $\frac{1}{16}F$?

5. Calculate the gravitational force between Earth and the Sun given:

$$r = 1.47 \times 10^{11}\,\mathrm{m}$$

6. Two masses $m_1 = 200\,\mathrm{kg}$ and $m_2 = 50\,\mathrm{kg}$ attract each other with force $F = 1.0 \times 10^{-6}\,\mathrm{N}$. Find the distance between them.

7. A satellite orbits Earth in a circular orbit of radius $7.00 \times 10^{6}\,\mathrm{m}$ from Earth's center. Find its orbital speed.